

1 In this experiment, you will investigate the phase difference between the oscillations of two mass–spring systems.

- (a) • Assemble the apparatus as shown in Fig. 1.1.

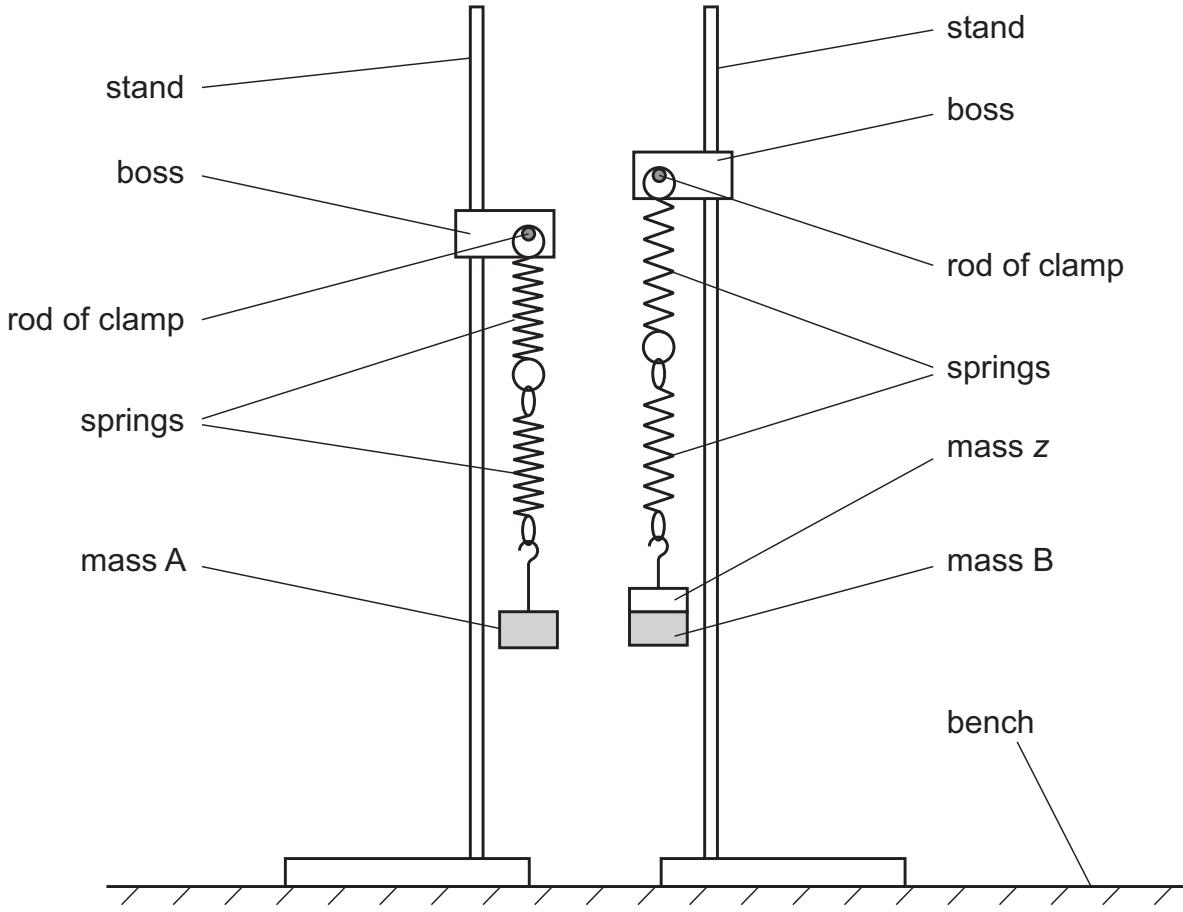


Fig. 1.1

- Mass A and mass B are each 200 g.
- Add a mass z of 40 g to mass B.

Record the value of z .

$z = \dots\dots\dots$ g

- M is given by $M = 200\text{ g} + z$.

Calculate M .

$M = \dots\dots\dots$ g

- Pull both A and B down a short distance and release them together. Observe the oscillations. A and B initially oscillate in phase (both moving up and down together), then their oscillations go out of phase and then become in phase again.
- The time from A and B oscillating in phase to the next time they oscillate in phase is P .

Measure and record P .

$P = \dots\dots\dots$ [2]

- (b) Change z and determine P . Repeat until you have six sets of values of z and P .

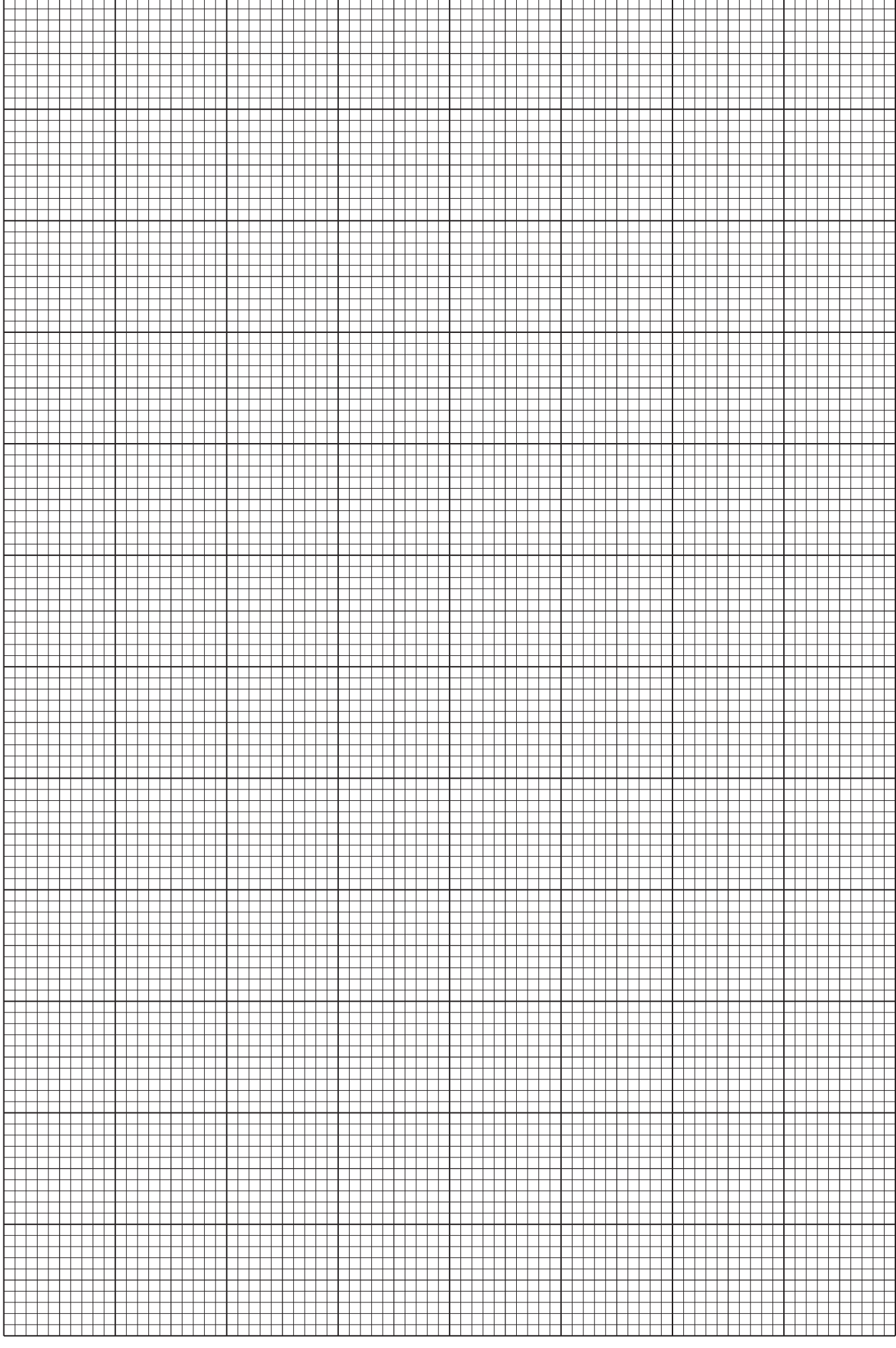
Record your results in a table. Include values of M , $\frac{1}{\sqrt{M}}$ and $\frac{1}{P}$ in your table.

[10]

- (c) (i) Plot a graph of $\frac{1}{P}$ on the y -axis against $\frac{1}{\sqrt{M}}$ on the x -axis. [3]
(ii) Draw the straight line of best fit. [1]
(iii) Determine the gradient and y -intercept of this line.

gradient = $\dots\dots\dots$

y -intercept = $\dots\dots\dots$ [2]



- (d) It is suggested that the quantities P and M are related by the equation

$$\frac{1}{P} = \frac{a}{\sqrt{M}} + b$$

where a and b are constants.

Use your answers in (c)(iii) to determine the values of a and b .
Give appropriate units.

$a = \dots\dots\dots$

$b = \dots\dots\dots$ [2]

[Total: 20]

You may not need to use all of the materials provided.